

Simple WATER TEMPERATURE INDICATOR And CONTROLLER Using Arduino

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If you wish to monitor and control the temperature of water, or any other liquid for that matter, this proposed add-on module can be of help. It will let you set the maximum temperature to which the water needs to be heated, display the temperature at any stage of the heating process, give indications through a buzzer and LED on reaching the desired temperature and switch the heater off.

Most storage-type water geysers available in market for home use have such features, but some water heaters—especially rod type water heaters—do not provide these facilities. They keep on heating water till they are switched off and thus consume lots of electricity, unnecessarily, and also heat the water much more than desired.

Block diagram of this simple project is shown in Fig. 1 and its circuit diagram in Fig. 2. The project is built around water temperature sensor probe DS18B20, a 4-digit multiplex 7-segment display, Arduino Uno board having microcontroller ATmega328, opto-couple MCT2E, a potentiometer (pot), LEDs, a mini buzzer, and relay.

The water temperature sensed by the probe can be seen on the 4-digit display. You can set the required water temperature using the pot and switch on the relay to start heating water. When the water temperature reaches the set level, the relay will automatically turn off and heating of the water will stop.

The one-wire sensor probe

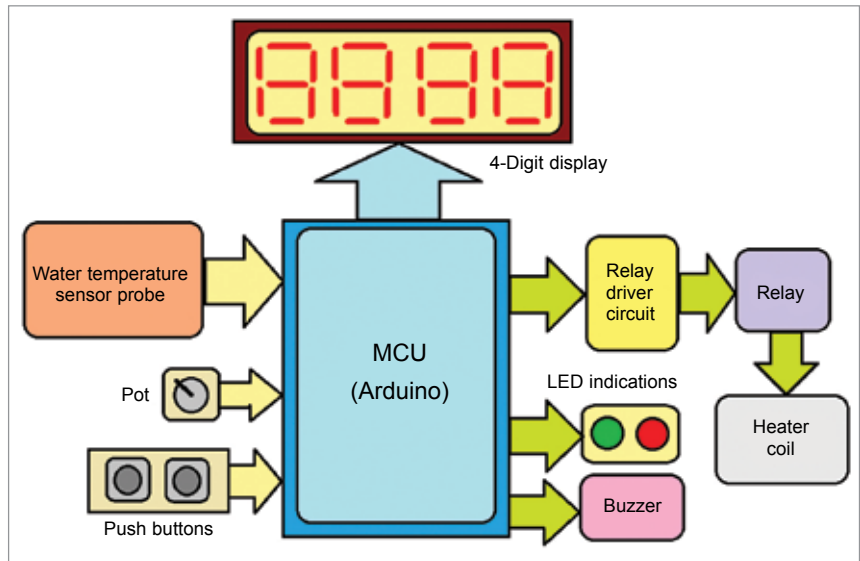


Fig. 1: Block diagram of the water temperature indicator and controller

DS18B20 from Dallas recommended for the project senses the water temperature and gives direct digital value of temperature reading using a special one-wire protocol. Pot VR1 is used to set the required water temperature. DIS1 is a 4-digit multiplex 7-segment display that helps set the required water temperature and know the temperature at any stage. LED1 and LED2 are heater on/off indicators. The buzzer produces 'beep' sound on reaching the set water temperature. Relay driver circuit (built around MCT2E) provides isolation and protection besides driving the relay.

The 10k pot (VR1) is connected between 5V and GND pins of Arduino board with its central slider terminal connected to analogue input pin A0. This gives 0 to 5V analogue input to Arduino board. Push-button

switches S1 and S2 are connected to pins A1 and A2 such that when a switch is pressed, the corresponding pin is grounded and gives logic 0 input. The two LEDs (red and green) are connected to pins A3 and A4 with their cathodes connected to ground through 330-ohm current-limiting resistors.

Pin A5 of Arduino Board1 is connected to anode pin of the LED in opto-coupler MCT2E through 330-ohm current limiting resistor R1. Output of the transistor in the opto-coupler drives the relay. A reverse freewheeling diode is connected across relay coil.

Mini buzzer PZ1 is connected to digital pin D0 of the board, with its + input connected to pin D0 and - input connected to ground. Sensor probe DS18B20 has three interfacing